



Department of Artificial Intelligence & Machine Learning
Academic Year 2025-2026

Name: Archit Vaishnav Tiwari

Roll no.: A121. Div: A2. Batch:A2-2.

Sap Id: 60017240174. Date: 30th July 2026.

Experiment No. 2

AIM: To implement the following Point Processing techniques:

1. Image Negative
2. Contrast Stretching
3. Thresholding
4. Gray-level Slicing.
5. Log Transformation, Power Law Transformation

THEORY:

The principle objective of image enhancement is to process an image so that the result is more suitable than the original image for a specific application. Image enhancement approaches fall into two broad categories:

Spatial domain methods - which involve direct manipulation of pixels in an image

Frequency domain methods - which are based on modifying the Fourier transform of an image.

Point processing techniques are among the simplest of all image enhancement techniques. These techniques are basically zero memory operations where, every pixel is operated individually without the use of neighbouring pixels.

The values of pixels, before and after processing, will be called r and s , respectively. These values are related by an expression of the form

$$s = T(r)$$

where T is a transform function, it maps the pixel value r in to a pixel values.

1. Image Negatives



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The negative of an image with gray levels in the range $\{0, L-1\}$ is obtained by using the negative transformation given by the expression,

$$s = (L - 1) - r$$

where,

s = output gray level.

r = input gray level.

$L - 1$ = Highest gray level present in the image.

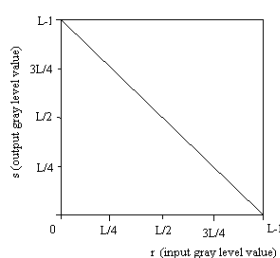


Fig 1. Negative Image Transformation

This type of processing is particularly suited for enhancing white or gray detail embedded in the dark regions of an image, especially when the black areas are dominant in size.

2. Contrast Stretching

Low - contrast images can result from poor illumination, lack of dynamic range in the imaging sensor or wrong setting of a lens aperture during image acquisition. Contrast stretching is employed to increase the dynamic range of the gray levels in the image being processed.

The figure illustrates the form of the transformation function for contrast stretching:

The locations of points (r_1, s_1) and (r_2, s_2) control the shape of the transformation and thus the contrast of the output image.

If $r_1 = s_1$ and $r_2 = s_2$, the transformation is a linear function that produces no changes in the gray levels.



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If $r_1 = r_2$, $s_1 = 0$ and $s_2 = L-1$, the transformation becomes a thresholding function that creates a binary image.

Intermediate values of (r_1, s_1) and (r_2, s_2) produce various degrees of spread in the gray levels of the output image, thus affecting its contrast.

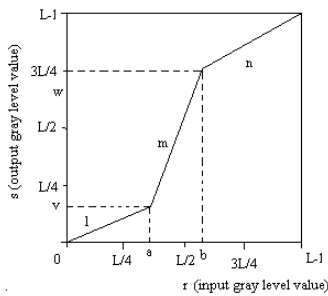


Fig 2. Contrast Stretching

In general, $r_1 \leq r_2$ and $s_1 \leq s_2$ is assumed to make the function single valued and monotonically increasing which preserves the order of gray levels thus preventing the creation of intensity artefacts in the processed image.

3. Thresholding

The input to a thresholding operation is typically a grayscale image. In the simplest implementation, the output is a binary image representing the segmentation. Black pixels correspond to background and white pixels correspond to foreground (or *vice versa*).

For example, a simple mapping function is defined by the thresholding function. In contrast stretching if $a = b$, $v = 0$ & $w = L - 1$, the transformation function is such that ,

$$s = 0 \quad \text{for } r < a$$

$$L - 1 \quad \text{for } r \geq a$$

The figure below shows the transformation function:



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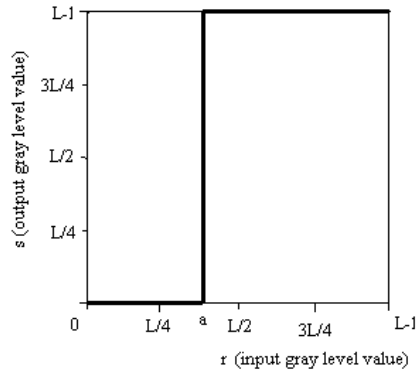


Fig 3. Thresholding

In simple implementations, the segmentation is determined by a single parameter known as the *intensity threshold*. In a single pass, each pixel in the image is compared with this threshold. If the pixel's intensity is higher than the threshold, the pixel is set to, say, white in the output. If it is less than the threshold, it is set to black.

4. Gray Level Slicing

Highlighting a specific range of gray levels in an image is often desired. Gray level slicing has two basic approaches for enhancement:

One is to display a high value for all gray levels in the range of interest and a low value for all other gray levels. This transformation produces a binary image.

Another approach brightens the desired range of gray levels but preserves the background and gray level tonalities in the image.

gray level slicing without background

$$s = L-1 \quad a \leq r < b$$

$$s = 0 \quad \text{otherwise}$$

gray level slicing with background

$$s = L-1 \quad a < r \leq b$$

$$s = r \quad \text{otherwise}$$

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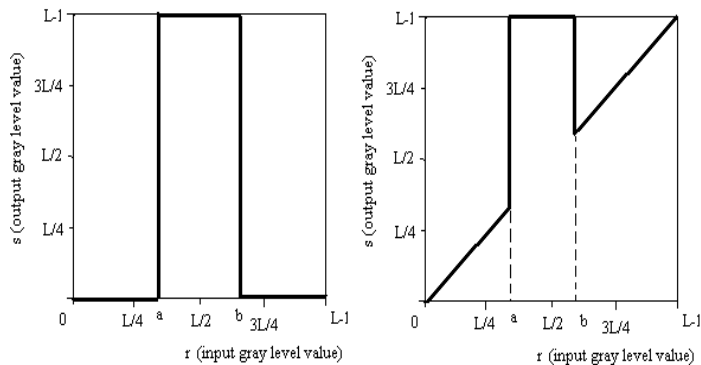


Fig 4. The gray level slicing operation (left), and with background operation (right).

5. Dynamic Range Compression (Log Transformation)

The log transformations can be defined by this formula

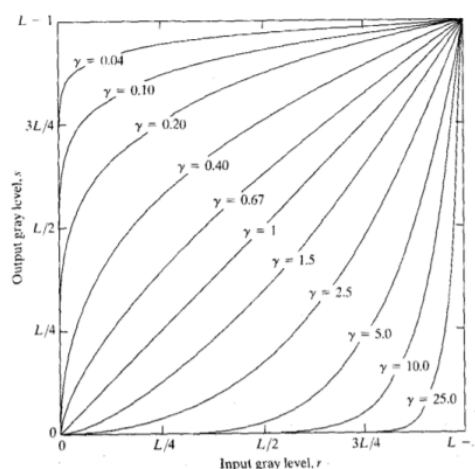
$$s = c \log(r + 1)$$

Where s and r are the pixel values of the output and the input image and c is a constant. The value 1 is added to each of the pixel value of the input image because if there is a pixel intensity of 0 in the image, then $\log(0)$ is equal to infinity. So 1 is added, to make the minimum value at least 1.

During log transformation, the dark pixels in an image are expanded as compare to the higher pixel values. The higher pixel values are kind of compressed in log transformation.

Power Law Transformation:

$$S=r^\gamma$$





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Code:

(Students Should Type code here)

```
import cv2
import numpy as np
from matplotlib import pyplot as plt

img = cv2.imread(r"C:\Users\djsce.student\Desktop\Cameraman.webp")

if img is not None:
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    plt.imshow(img)
    plt.title("Image of a Cameraman")
    plt.axis("off")
    plt.show()
else:
    print("Error")
```

Image of a Cameraman



```
L = img.max()
img_neg = L - img
plt.imshow(img_neg)
plt.title("Negative Image")
plt.axis("off")
plt.show()
```

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Negative Image



```
print(L)
```

```
255
```

```
r_min, r_max = np.min(img), np.max(img)

#r_min, r_max = 20, 220
contrast_stretch = (((img-r_min)/(r_max-r_min))*255).astype(np.uint8)

plt.subplot(1, 2, 1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(contrast_stretch, cmap='gray'); plt.title('Contrast Str')
plt.show()
```

Original



Contrast Stretched



```
threshold_value = 128
_, binary_img = cv2.threshold(img, threshold_value, 255, cv2.THRESH_BINARY)

plt.subplot(1, 2, 1); plt.imshow(img, cmap='gray'); plt.title('Grayscale'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(binary_img, cmap='gray'); plt.title(f'Threshold: {thre')
plt.show()
```


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Grayscale



Threshold: 128



```
lower, upper = 80, 200
sliced = np.where((img >= lower) & (img <= upper), 255, 0).astype(np.uint8)
```

```
plt.subplot(1, 2, 1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(sliced, cmap='gray'); plt.title('Gray-level Slicing');
plt.show()
```

Original



Gray-level Slicing



```
c = 255/np.log(1+np.max(img))
log_transformed = c*(np.log(img+1))
log_transformed = np.array(log_transformed, dtype=np.uint8)
```

```
plt.subplot(1, 2, 1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(log_transformed, cmap='gray'); plt.title('Log Transform');
plt.show()
```




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```
C:\Users\djsce.student\AppData\Local\Temp\ipykernel_23884\893444732.py:2: RuntimeWarning  
log_transformed = c*(np.log(img+1))  
C:\Users\djsce.student\AppData\Local\Temp\ipykernel_23884\893444732.py:3: RuntimeWarning  
log_transformed = np.array(log_transformed, dtype=np.uint8)
```

Original



Log Transformed



```
np.max(img)
```

255



```
gamma = 0.5  
c = 255/np.max(img)**gamma  
powerlaw_img = c*(img**gamma)  
powerlaw_img = np.array(powerlaw_img, dtype=np.uint8)
```

```
plt.subplot(1, 2, 1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')  
plt.subplot(1, 2, 2); plt.imshow(powerlaw_img, cmap='gray'); plt.title('Power Law'); plt.  
plt.show()
```

Original



Power Law





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Post Lab Session: (Answer the following questions)

1. Take an image given in your lab session. Manually apply Digital Negative, contrast stretching, grey level slicing, Thresholding and power law transformation to the same.

30/07/2026 EXP02

ARCHIT VAISHNAV
 A121, 60017240174
 A2-L

* AIM: To implement Image negative, contrast-stretching, Thresholding, Grey level slicing, log transformation, Power Law Transform.

→ $Img =$

82	14	37	43	52
53	22	29	46	59
34	55	18	43	23
7	39	44	47	09
41	83	29	31	11

(i) Image Inverse:
 $S = (L-1) - r$ L: Highest-grey level
 $L = 64$ i.e. 2^6 as highest num is 59

∴ inv $Img =$

31	49	26	18	11
10	41	40	17	44
29	8	45	20	40
56	24	19	16	54
22	30	34	32	52

(ii) Contrast-Stretching.
 $S = \text{round} \left(\frac{S_{max} - S_{min}}{r_{max} - r_{min}} (r - r_{min}) + S_{min} \right)$

$S = \text{round} \left(\frac{(63 - 0)}{59 - 7} (r - 7) \right) = \text{round} (1.2115 (r - 7))$

∴ For $r = 7$, $1.2115 (7 - 7) = 0$
 $r = 8$, 1.2115

Output matrix:

30	8	36	46	55
56	18	19	17	45
33	58	13	41	19
0	8	45	20	40
41	31	37	29	2



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(iii) Thresholding:

$$S = \begin{cases} 63 & x \geq 32 \\ 0 & x < 32 \end{cases}$$

$$O = \begin{bmatrix} 63 & 0 & 63 & 63 & 63 \\ 63 & 0 & 0 & 63 & 63 \\ 63 & 63 & 0 & 63 & 0 \\ 0 & 63 & 63 & 63 & 0 \\ 63 & 63 & 0 & 0 & 0 \end{bmatrix}$$

(iv) Grey level slicing:

$$S = \begin{cases} 63 & 30 \leq r \leq 45 \\ 0 & \text{otherwise} \end{cases}$$

$$O = \begin{bmatrix} 63 & 0 & 63 & 63 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 63 & 0 & 0 & 63 & 0 \\ 0 & 63 & 63 & 0 & 0 \\ 63 & 63 & 0 & 63 & 0 \end{bmatrix}$$

(v) Log Transformation: $S = c \cdot \log_{10}(1+r)$.

$$\therefore 63 = c \cdot \log_{10}(1+63) = c \cdot \log_{10}(64) = 34.88.$$

$$S = \text{round}(34.88 \cdot \log_{10}(1+r))$$

$$O = \begin{bmatrix} 53 & 41 & 55 & 58 & 66 \\ 60 & 48 & 48 & 58 & 62 \\ 54 & 61 & 45 & 57 & 48 \\ 51 & 56 & 58 & 54 & 35 \\ 57 & 53 & 52 & 53 & 38 \end{bmatrix}$$

(vi) Gamma Transformation:

$$S = c \cdot r^r$$

$$S = \text{round}\left(\frac{r^2}{63}\right)$$

$$O = \begin{bmatrix} 16 & 3 & 22 & 32 & 43 \\ 45 & 8 & 8 & 34 & 55 \\ 16 & 48 & 5 & 24 & 8 \\ 1 & 24 & 91 & 35 & 1 \\ 27 & 17 & 13 & 15 & 2 \end{bmatrix}$$

FOR EDUCATIONAL USE



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Conclusion:

The experiment was successfully carried out to implement various point processing techniques, including Image Negative, Contrast Stretching, Thresholding, Gray-level Slicing, Log Transformation, and Power Law Transformation. These techniques demonstrated how individual pixel values can be modified to enhance image quality, improve contrast, highlight specific intensity ranges, and reveal hidden details. The experiment helped in understanding the practical applications of point processing in image enhancement and digital image analysis.